

**VIT®****Vellore Institute of Technology**
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CHENNAI

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Final Assessment Test(FAT) - NOV/DEC 2025

Programme	B.Tech.	Semester	Fall Semester 2025-26
Course Code	BCSE324L	Faculty Name	Prof. Avuthu Avinash Reddy
Course Title	Foundations of Blockchain Technology	Slot	G1+TG1
		Class Nbr	CH2025260100726
Time	3 hours	Max. Marks	100

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

- CO1: Understand Blockchain ecosystem and its services in real world sceneries
- CO2: Apply and Analyze the requirement of Distributed Ledger Technology and Smart Contract
- CO3: Design and Demonstrate end-to-end decentralized applications
- CO4: Acquaint the protocol and assess their computational requirements

Section - I**Answer all Questions (7 × 10 Marks)**

01. i) A cold chain logistics company employs a blockchain network to monitor vaccine shipments from manufacturing facilities to hospitals. Each transaction captures critical parameters such as container temperature, timestamp, and location. During deployment, however, certain nodes occasionally report slight variations in data for the same shipment block. Discuss the key components of blockchain architecture that ensure secure and consistent data recording across all nodes in this cold-chain system. Illustrate how these components interact to maintain data accuracy and prevent inconsistencies across the network. (5 Marks)
- ii) In the same blockchain system, auditors regularly need to verify that temperature readings collected from IoT sensors remain unaltered after being recorded. Due to network latency, block synchronization delays sometimes result in mismatched Merkle roots among nodes. Describe how the Merkle tree structure facilitates data integrity verification during audits. Explain the impact of altering a single leaf hash (representing one temperature record) on the Merkle root, and how this mechanism helps in detecting data tampering. (5 Marks)
- [10] (CO1/K4)
02. A new cryptocurrency exchange wants to allow users to prove they own enough tokens to execute a transaction or participate in staking without revealing their exact wallet balance or transaction history. To maintain privacy, security, and trust on the platform, the system plans to implement Zero-Knowledge Proofs (ZKPs).
- i) Discuss the three fundamental properties of a Zero-Knowledge Proof in this context. (3 Marks)
- ii) Differentiate between Interactive and Non-Interactive ZKPs, and justify which type would be more suitable for this cryptocurrency platform. (3 Marks)
- iii) Compare ZK-SNARK and ZK-STARK in terms of trusted setup, proof size, scalability, verification speed. (4 Marks)
- [10] (CO2/K3)
03. A blockchain-based Medical Treatment Smart Contract is being developed for managing the digital workflow between hospitals, pharmacies, and patients. In this system, a doctor initiates the treatment by recording a diagnosis and creating a prescription. The pharmacy verifies the prescription and supplies the prescribed medicines. The patient can only view the prescription details and confirm receipt of medicines. The contract progresses through three lifecycle stages: a) Issued- Prescription created by the doctor. b) Dispensed- Pharmacy validates and dispenses the medicine. c) Acknowledged- Patient confirms receipt, marking completion. The system must ensure that only authorized roles can execute their respective functions, and once a stage is completed, the workflow cannot revert to a previous state.
- i) Identify the two smart contract design patterns applicable in this system and describe their purposes. (2 Marks)
- ii) Explain how user permissions can be managed so that only authorized participants can execute specific contract functions. (4 Marks)

- iii) Identify and Describe how the prescription process lifecycle can be modeled within the contract. Include the main stages and possible transitions. (4 Marks) [10] (CO2/K4)
04. A renewable energy firm, GreenVolt Pvt. Ltd., initially operates under a Centralized-Distributed (Ce-Di) organizational model. The company's regional branches handle local solar installations, but all financial and strategic decisions must be approved by headquarters. As GreenVolt expands across multiple countries, coordination delays and trust issues arise between branches. The management is considering moving to a Decentralized-Distributed (De-Di) model supported by blockchain. (5 Marks)
- Explain how moving from a Ce-Di to a De-Di model can improve coordination and trust in GreenVolt. (5 Marks)
 - Describe one challenge GreenVolt might face after decentralizing decision-making. Also, suggest how blockchain and smart contracts can help manage GreenVolt's decentralized operations. (5 Marks)
- [10] (CO3/K4)
05. A research consortium named DataVerse Labs is exploring blockchain-based decentralized solutions to enhance data analytics and forecasting services. They are evaluating the integration of Augur (a decentralized prediction market protocol) and Golem (a decentralized computation protocol). In their proposed model:
- Augur will be used to create markets for predicting climate change impacts, agricultural yields, and energy demand.
 - Golem will provide decentralized computing power to process massive climate datasets, enabling participants to rent computing resources in exchange for GNT tokens.
 - The combined system aims to reduce dependency on centralized cloud providers while rewarding participants for both accurate predictions and computing contributions.
- Illustrate how Augur and Golem represent different services within the blockchain ecosystem and their relevance to DataVerse Labs' use case. (5 Marks)
 - Analyze how Distributed Ledger Technology (DLT) and Smart Contracts are utilized in both protocols to ensure data integrity, trust, and automation. (5 Marks)
- [10] (CO4/K2)
06. A decentralized ride-sharing blockchain platform, RideChain, aims to eliminate intermediaries like cab companies.
- Leaders: Platform developers who validate critical blocks.
 - Core members: Regional fleet operators managing driver groups.
 - Participants: Individual drivers who add trip records.
 - Users: Passengers who book rides through smart contracts.
 - Third-party services: Insurance, map APIs, and payment gateways.
- Explain the roles and interactions of all five ecosystem components in maintaining network reliability and transparency. (5 Marks)
 - Analyse how integration of third-party APIs and smart contracts enhances functionality but introduces security and trust challenges. (5 Marks)
- [10] (CO1,2/K1)
07. A fintech company wants to set up a High-Performance Computing (HPC) cluster to run real-time risk analysis and fraud detection models. To ensure data integrity, auditability, and secure coordination across the cluster, they decide to integrate blockchain technology into the HPC environment. The cluster will consist of heterogeneous nodes distributed across multiple locations. The company also plans to simulate workloads before full deployment to identify performance bottlenecks.
- Design a cluster deployment strategy for the HPC nodes to ensure efficient blockchain integration. (5 Marks)
 - Discuss potential challenges such as node heterogeneity, latency, and synchronization. (2 Marks)
 - Suggest a mock workload to test the cluster and blockchain system. Identify what metrics should be measured to evaluate performance, scalability, and reliability. (3 Marks)
- [10] (CO4/K4)

Section - II

Answer all Questions (2 × 15 Marks)

08. A smart energy trading platform uses a blockchain network for recording peer-to-peer (P2P) electricity transactions between consumers and producers. The platform must ensure every energy trade is verified and recorded even when some IoT-enabled smart meters behave unreliably. During field trials,
- Certain meters broadcast duplicate transaction blocks.
 - Random communication delays cause consensus stalls.
 - Some nodes drop messages intentionally to disrupt the network.

i) Explain the Byzantine Fault Tolerance mechanism in detecting and managing faulty or malicious meter nodes. (5 Marks)

ii) Describe how the Garay backbone model defines liveness and persistence, and illustrate how these properties maintain correct ledger growth. (5 Marks)

iii) Compare how RLA model enhancements and AAP protocol can be used together to ensure asynchronous consensus under network delay. Recommend which combination is best suited for energy trading systems. (5 Marks)

[15] (CO1/K2)

09. a. A global tourism company plans to build a decentralized travel platform to manage bookings, digital payments, and real-time traveler data across multiple countries. The team is evaluating Blockchain, Hashgraph, Holochain, and DAG-based systems to ensure scalability, fairness, transparency, and low latency in transactions.

i) Analyze which of these distributed ledger technologies would be most effective for this tourism platform, and justify your selection. (4 Marks)

ii) Compare the consensus mechanisms used in each technology in terms of speed, scalability, energy efficiency, and fairness. (4 Marks)

iii) Recommend which consensus mechanism aligns best with the tourism system's requirements and explain why. (2 Marks)

b. A state electricity department is implementing a smart energy management system to record power generation, transmission, and consumption data across multiple cities. The system involves Consumer billing and renewable energy credit tracking, Smart meter device registry with authorized energy provider list, Power generation plants, and Regional transmission units. Identify which parts of this electricity ecosystem should use a public ledger, a private ledger, and a registry, and justify your selection based on data sensitivity, accessibility, and trust among the participants. (5 Marks)

[15] (CO2/K3)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)

